

computer industry curve of Moore's Law, because it is based on improvements in hardware. WiFi devices become cheaper and more sophisticated every year, just like personal computers (but unlike most telecommunications services). They are standards-based components sold in a competitive market, at volumes that allow for economies of scale. Those new devices become part of the network as soon as users purchase and install them. Capital investment is spread among users, rather than shouldered upfront by a network operator. By contrast, 3G services can only be deployed after service providers build and upgrade expansive network infrastructure and proprietary hardware.

Limitations in WiFi devices are being addressed through market forces. For example, first-generation WiFi equipment has a relatively weak built-in security mechanism known as Wireless Equivalent Privacy (WEP). For users concerned about security, such as enterprises, third parties and hardware vendors quickly developed supplemental security solutions that integrated with standards-based WiFi deployments. Meanwhile, an enhanced security standard, 802.1x, was recently ratified by the IEEE.

A key to the success of WiFi is that it uses a different business model than traditional telecommunications and broadband services. Because the network grows incrementally with every new access point and every device capable of receiving WiFi signals, there is no need for incentives to convince a monopoly service provider to build out expensive infrastructure. No one needs to predict what the killer applications of the technology will be, because users will find them on their own. With a licensed service, the network operator must invest in delivering services in the hope that customers will pay enough to recoup that investment. With WiFi, services grow bottom-up through market forces.

There are many things WiFi cannot do. For example, as a short-range LAN technology, it can't provide universal coverage over a large area, and it isn't designed for mobile scenarios such